## PRACTICE QUESTIONS FOR PRESENTATION OF QUALITATIVE DATA

# EXERCISE 1.2

**1.** **Career Changes** A survey asked if people would like to spend the rest of their careers with their present employers. The results are shown.

| Answer | Number of people |
|---|---|
| Yes | 660 |
| No | 260 |
| Undecided | 80 |

(a) Construct a bar chart for the data and analyze the results.
(b) Construct a pie chart for the data and analyze the results.

**2.** **Spending of College Freshmen** The average amounts spent by college freshmen for school items are shown.

| | |
|---|---|
| Electronics/computers | $728 |
| Dorm items | 344 |
| Clothing | 141 |
| Shoes | 72 |

Source: National Retail Federation.

**(a)** Construct a bar graph for the data. **(b)** Construct a pie graph for the data.

**3.** **Housing Arrangements** A questionnaire on housing arrangements showed this information obtained from 25 respondents. Construct a frequency distribution for the data (H = house, A apartment, M = mobilehome, C = condominium).

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| H | C | H | M | H | A | C | A | M |
| C | M | C | A | M | A | C | C | M |
| C | C | H | A | H | H | M | | |

**(a)** Construct a bar graph for the data. **(b)** Construct a pie graph for the data.

**4.** **Trust in Internet Information** A survey was taken on how much trust people place in the information they read on the Internet. A trust in everything they read, M trust in most of what they read, H trust in about one-half of what they read, S trust in a small portion of what they read. (Based on information from the *UCLA Internet Report*.)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| M | M | M | A | H | M | S | M | H | M |
| S | M | M | M | M | A | M | M | A | M |
| M | M | H | M | M | M | H | M | H | M |
| A | M | M | M | H | M | M | M | M | M |

(a) Construct categorical & relative categorical frequency distribution for the data and interpret the results.
(b) Construct a bar graph for the data and a pie graph for the data.

**5.** **How People Get Their News** The Brunswick Research Organization surveyed 50 randomly selected individuals and asked them the primary way they received the daily news. Their choices were via newspaper (N), television (T), radio (R), or Internet (I).

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| N | N | T | T | T | I | R | R | I | T |
| I | N | R | R | I | N | N | I | T | N |
| I | R | T | T | T | T | N | R | R | I |
| R | R | I | N | T | R | T | I | I | T |
| T | I | N | T | T | I | R | N | R | T |

(a) Construct categorical & relative categorical frequency distribution for the data and interpret the results.
(b) Construct a bar graph for the data and a pie graph for the data.

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# PRACTICE QUESTIONS FOR PRESENTATION OF QUANTITATIVE DATA

## EXERCISE 1.3

**1.** **NFL Salaries** The salaries (in millions of dollars) for 31 NFL teams for a specific season are given in this frequency distribution

| Class limits | Frequency |
|---|---|
| 39.9–42.8 | 2 |
| 42.9–45.8 | 2 |
| 45.9–48.8 | 5 |
| 48.9–51.8 | 5 |
| 51.9–54.8 | 12 |
| 54.9–57.8 | 5 |

Source: NFL.com

Construct a histogram, a frequency polygon, and an ogive for the data; and comment on the shape of the distribution.

**2.** **How Quick Are Older Dogs?** The animal trainer selected a group of dogs who were much older than the first group and measured their reaction times to the same stimulus. Construct a histogram, a frequency polygon, and an ogive for the data.

| Class limits | Frequency |
|---|---|
| 2.3–2.9 | 1 |
| 3.0–3.6 | 3 |
| 3.7–4.3 | 4 |
| 4.4–5.0 | 16 |
| 5.1–5.7 | 14 |
| 5.8–6.4 | 4 |

**3.** **Presidential Vetoes** The number of total vetoes exercised by the past 20 Presidents is listed below. Use the data to construct a grouped frequency distribution and a cumulative frequency distribution with 5 classes. Construct a histogram, a frequency polygon, and an ogive for the data.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 44 | 39 | 37 | 21 | 31 | 170 | 44 | 635 | 30 | 78 |
| 42 | 6 | 250 | 43 | 10 | 82 | 50 | 181 | 66 | 37 |

**4.** **Grams per Food Serving** The data shown are the number of grams per serving of 30 selected brands of cakes. Use the data to construct a frequency distribution with 6 classes, a histogram, a frequency polygon, and an ogive.

| | | | | | |
|---|---|---|---|---|---|
| 32 | 47 | 51 | 41 | 46 | 30 |
| 46 | 38 | 34 | 34 | 52 | 48 |
| 48 | 38 | 43 | 41 | 21 | 24 |
| 25 | 29 | 33 | 45 | 51 | 32 |
| 32 | 27 | 23 | 23 | 34 | 35 |

Source: *The Complete Food Counts.*

**5.** **Weights of the NBA's Top 50 Players** Listed are the weights of the NBA's top 50 players. Construct a grouped frequency distribution and a cumulative frequency distribution.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 240 | 210 | 220 | 260 | 250 | 195 | 230 | 270 | 325 | 225 |
| 165 | 295 | 205 | 230 | 250 | 210 | 220 | 210 | 230 | 202 |
| 250 | 265 | 230 | 210 | 240 | 245 | 225 | 180 | 175 | 215 |
| 215 | 235 | 245 | 250 | 215 | 210 | 195 | 240 | 240 | 225 |
| 260 | 210 | 190 | 260 | 230 | 190 | 210 | 230 | 185 | 260 |

Source: www.msn.foxsports.com

Construct a histogram, a frequency polygon, and an ogive for the data.

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